NUI Galway
OÉ Gaillimh

Semester II Examinations 2016/2017

Exam Code(s)	3BS9, 3BPT1, 4BS2, 4BME1, 1CSD1
Exam(s)	Third & Fourth Science; Fourth Maths and Education; Data Analytics
Module(s)	Applied Statistics 2
Module Code(s)	ST312
Paper No	1
Repeat Paper	
External Examiner	Prof. S. Wilson
Internal Examiner	Prof. J. P. Hinde

Instructions:

Answer ALL 5 Questions.

Duration	2 Hours
No. of Pages	12 Pages
Department(s)	Statistics
Course Co-ordinators(s)	

Requirements:

Release to Library:	Yes
Statistical Tables/ Log Tables	Yes
Other Materials	Standard (non-programmable) calculator is allowed

Question 1: [Total marks 15]

To investigate whether it helps smokers to quit smoking if they take antidepressants, one study used 429 men and women who were 18 or older and had smoked 15 cigarettes or more per day for the previous year. The subjects were highly motivated to quit and in good health. They were assigned to one of two groups:

- one group took 300 mg daily of buproprion, an antidepressant that has the brand name Zyban;
- the other group did not take an antidepressant.

At the end of a year, the study observed whether each subject had successfully abstained from smoking or had relapsed.

(a) Is this an observational or experimental study? Explain your answer. *(2 marks)*

(b) Identify the responses and explanatory variables, the treatments, and the experimental units. *(4 marks)*

(c) How should the researchers assign the subjects to the two groups? Why? Explain the advantages of your proposed approach. *(3 marks)*

(d) Without knowing more about this study, what would you identify as potential problems with the study design? *(3 marks)*

(e) If the researchers believed that there could be gender differences in the effect of antidepressants on quitting smoking, how would you suggest modifying the study? *(3 marks)*

Question 2: [Total marks 15]

The data below give the elasticity of billiard balls made under three different conditions. Ten batches of melted plastic were prepared. Each batch was divided into three equal portions. One portion was chosen at random and set aside as a control. A second portion was chosen at random from the remaining two, and was mixed with additive A. The third portion was mixed with additive B. Elasticity was measured on a scale from 0 to 100, with the higher number representing greater elasticity. (Higher elasticity is considered more desirable.)

	Batch									
	1	2	3	4	5	6	7	8	9	10
Control	51	45	49	66	53	41	58	56	60	63
Additive A	75	89	73	84	66	85	73	71	78	65
Additive B	39	43	51	34	54	43	42	41	37	44

(a) List the factor of interest and any nuisance factors. *(2 marks)*

(b) What are the experimental units? *(1 mark)*

(c) Are there blocks in this design? If so, identify them. *(2 marks)*

(d) Why did the experimenters conduct the experiment in this way? *(2 marks)*

(e) Why did they chose the portions at random? *(2 marks)*

(f) Describe how a randomization test could be used to see if the additives have any effect on elasticity. State the null and alternative hypotheses being tested and describe (briefly) how such a test would be implemented including how the associated p-value would be calculated. What would be the advantage of doing this rather than a standard ANOVA F-test?
(No calculations are required.) *(6 marks)*

Question 3: [Total marks 35]

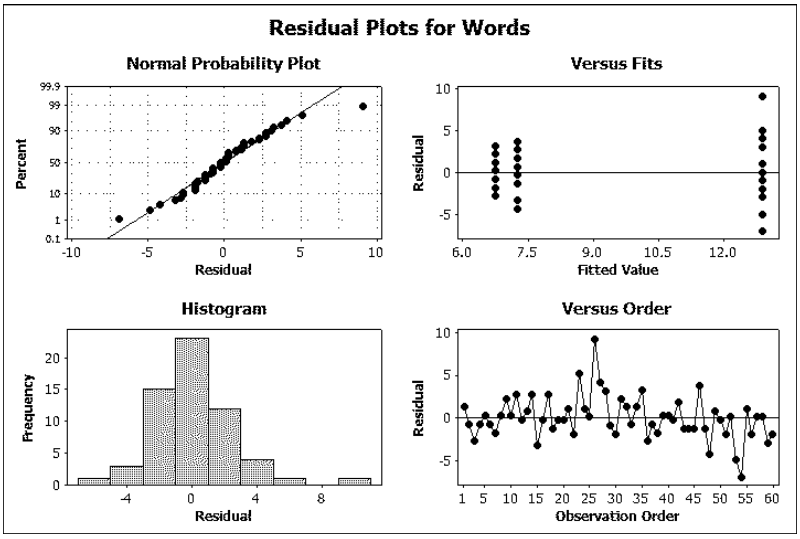
One theory regarding memory is that verbal material is remembered as a function of the degree to which is was processed when it was initially presented. As part of an experiment, Eysenck (1974) randomly assigned 60 subjects to one of three learning groups (20 to each). The Counting group was asked to read through a list of words and count the number of letters in each word. This involved the lowest level of processing. The Rhyming group was asked to read each word and think of a word that rhymed with it. The Adjective group was asked to give an adjective that could reasonably be used to modify each word in the list. None of the groups was told they would later be asked to recall the items. After the subjects had gone through the list of 27 items three times they were asked to write down all the words they could remember and the word count was used as a measure of the effectiveness of the memory strategy.

(a) Using the Minitab output below, test the hypothesis that the method of learning has no effect on effectiveness. Clearly specify your null and alternative hypotheses, calculate appropriate F-statistic and use the tables at the end of this paper to perform a test at the 5% level. *(6 marks)*

```
One-way ANOVA: Words versus Process

Source   DF      SS      MS      F      P
Process   2  466.63  233.32    ***    ***
Error    57  365.30    6.41
Total    59  831.93

S = 2.532   R-Sq = 56.09    R-Sq(adj) = 54.55%

                              Individual 95% CIs For Mean Based on
                              Pooled StDev
Level       N    Mean  StDev  --------+---------+---------+---------+-
Adjective  20  12.900  3.538                           (----*---)
Counting   20   6.750  1.618  (----*----)
Rhyming    20   7.250  2.023    (----*----)
                              --------+---------+---------+---------+-
                                    7.5      10.0      12.5      15.0

Pooled StDev = 2.532
```

Question continues . . .

(b) Comment on the validity of the assumptions for this ANOVA based on the residuals plots below. If you have reservations on the validity, suggest what you might do to improve matters. *(4 marks)*



(c) Following on from the above ANOVA why does it make sense to perform a multiple comparisons procedure?
Use the following output to perform multiple comparisons. What are the *family-wise* and *individual* type I error rates of your procedure? Group the different learning processes accordingly. *(4 marks)*

```
Tukey 95% Simultaneous Confidence Intervals
All Pairwise Comparisons among Levels of Process

Individual confidence level = 98.05%


Process = Adjective subtracted from:

Process    Lower  Center   Upper  -------+---------+---------+---------+--
Counting  -8.075  -6.150  -4.225  (-----*------)
Rhyming   -7.575  -5.650  -3.725    (-----*------)
                                  -------+---------+---------+---------+--
                                       -6.0      -3.0       0.0       3.0


Process = Counting subtracted from:

Process   Lower  Center  Upper  -------+---------+---------+---------+--
Rhyming  -1.425   0.500  2.425                     (------*-----)
                                -------+---------+---------+---------+--
                                     -6.0      -3.0       0.0       3.0
```

(d) The researchers were interested in testing whether the Adjective approach was different from the other two. Define a suitable contrast to compare the score for the Adjective approach with the combined score for Rhyming and

Counting. Specify the null and alternative hypotheses being tested, obtain an estimate for the contrast statistic, calculate its standard error and the degrees of freedom for the test. Perform the test at the 5% significance level. *(6 marks)*

(e) In fact the experimental subjects were drawn from two different age groups (Older: between 55 and 65 years old; Younger: between 25 and 30 years old) with 30 from each age group. The random allocation to learning process assigned equal numbers of subjects to each method. What do you conclude from the two-way ANOVA output given below? *(5 marks)*

```
Two-way ANOVA: Words versus Age, Process

Source   DF       SS       MS      F      P

Age       1   26.667   26.667   4.41  0.040
Process   2  466.633  233.317  38.58  0.000
Error    56  338.633    6.047
Total    59  831.933

S = 2.459   R-Sq = 59.30%   R-Sq(adj) = 57.12%
```

(f) The model in the above analysis is just a main effects model. The tables below give the cell means (based on 10 observations per cell) and a partial table of interaction effects. Complete the entries in the interaction effects table.
If there were no interaction, what would be the values in this table? *(4 marks)*

Cell means	Older	Younger	Row Mean
Adjective	11.000	14.800	12.900
Counting	7.000	6.500	6.750
Rhyming	6.900	7.600	7.250
Column Mean	8.300	9.633	8.967

Interaction Effect	Older	Younger
Adjective	-1.23	A
Counting	B	-0.916
Rhyming	0.317	-0.317

(g) Calculate the Interaction Sum of Squares (remembering that there are 10 observations per cell) and its degrees of freedom. Hence, extend the two-way ANOVA table in (e) to include the interaction term, calculating the new Error SS, Error MS and effect F-statistics. Test the significance of the interaction? What do you conclude? *(6 marks)*

Question 4: [Total marks 10]

The following 2×2 table is from a randomised control trial of a new treatment for gout, *CureIt!*, against a placebo control. 20 gout patients were randomised to the two arms of the trial (10 to treatment and 10 to control) and reassessed 30 days later.

	Cured	Not Cured	Total
CureIt!	8	2	10
Control	3	7	10
Total	11	9	20

Compare the treatment and control groups using the following measures and state what the relevant null and alternative hypotheses would be. You do not need to perform the test, simply calculate the test statistics.

(i) Equality of cured proportions;

(ii) Relative Risk;

(iii) Odds-ratio.

(6 marks)

For small studies, such as this, standard large-sample results based on the normal distribution can be unreliable. Fisher's exact test is designed to deal with this problem and calculates a randomization p-value based on the observed table and table(s) with the same marginal totals that show stronger evidence of a treatment effect. List the table(s) that are more extreme than the above table. *(4 marks)*

Question 5: [Total marks 25]

In an insecticide trial flour beetles were sprayed with one of two different insecticides (A: DDT; B: benzene-hexachloride). Batches of insects were exposed to varying doses of the spray, measured in mg/10 cm^2. For each batch the number of insects and the number of insects killed after six days was recorded. In the following we are interested in modelling the probability of death using logistic regression models.

The output below is from a model exploring the relationship between the binary variable `Death` ($1 =$dead; $0 =$alive) and the `dose` level.

Fit 1: `Logistic Regression Table`

					Odds	95% CI	
Predictor	Coef	SE Coef	Z	P	Ratio	Lower	Upper
Constant	-1.15796	0.215507	-5.37	0.000			
dose	0.394755	0.0498604	7.92	0.000	1.48	1.35	1.64

```
Log-Likelihood = -349.012
Test that all slopes are zero: G = 75.745, DF = 1, P-Value = 0.000
```

(a) Write down the fitted model in both the logit and probability forms of the logistic regression model. *(3 marks)*

(b) Interpret the estimated slope for `dose`. *(2 marks)*

(c) Is there a statistically significant relationship between `Death` and `dose`? State the null and alternative hypotheses, the Wald statistic, associated p-value, and your conclusions.
What is another alternative way of testing this relationship? *(5 marks)*

(d) Use the fitted model to predict the probability of death at a dose of 5 mg/10 cm^2. *(2 marks)*

(e) Toxicologists often summarise the potency of an insecticide in terms of LD_{50}, the (lethal) dose required to give a 50% kill rate (i.e. a 0.5 probability of death). Use the fitted model to estimate the LD_{50}. *(3 marks)*

(f) Calculate a 95% confidence interval for the slope parameter and explain how this relates to the reported interval for the *Odds Ratio*. *(3 marks)*

Question continues . . .

The above analysis ignores possible differences between the two insecticides. To explore this two more models were fitted as shown below.

Fit 2: Logistic Regression Table

Predictor	Coef	SE Coef	Z	P	Odds Ratio	95% CI Lower	95% CI Upper
Constant	-3.07749	0.329430	-9.34	0.000			
dose	0.548365	0.0622624	8.81	0.000	1.73	1.53	1.96
insecticide							
B	2.82917	0.244921	11.55	0.000	16.93	10.48	27.36

```
Log-Likelihood = -258.748
Test that all slopes are zero: G = 256.273, DF = 2, P-Value = 0.000
```

Fit 3: Logistic Regression Table

Predictor	Coef	SE Coef	Z	P	Odds Ratio	95% CI Lower	95% CI Upper
Constant	-2.67625	0.346517	-7.72	0.000			
dose	0.464105	0.0669586	6.93	0.000	1.59	1.39	1.81
insecticide							
B	1.11782	0.678363	1.65	0.099	3.06	0.81	11.56
insecticide*dose							
B	0.532820	0.212971	2.50	0.012	1.70	1.12	2.59

```
Log-Likelihood = -254.624
Test that all slopes are zero: G = 264.520, DF = 3, P-Value = 0.000
```

(i) Use the output (specifically the G values) from the three model fits to produce an analysis of deviance tables for the effects of `dose`, `insecticide` and their interaction.
What do you conclude from this table? Justify you conclusion with a formal significance test. *(5 marks)*

(ii) Compare the effect on the odds ratio of a one-unit increase in dose for the two insecticides. *(2 marks)*

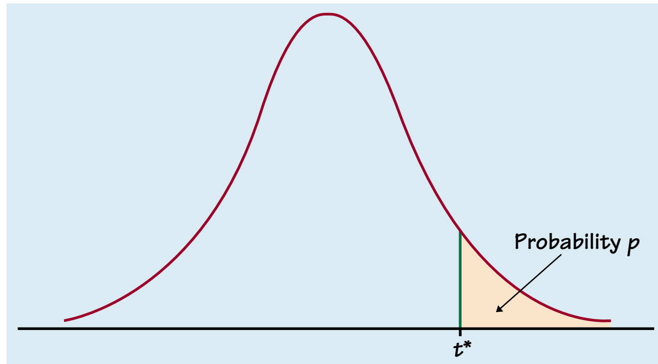


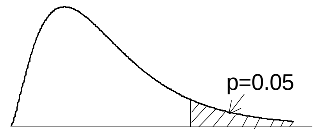
Table entry for p and C is the point t^* with probability p lying above it and probability C lying between $-t^*$ and t^*.

TABLE D ***t* distribution critical values**

	Tail probability p											
df	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	127.3	318.3	636.6
2	0.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60
3	0.765	0.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92
4	0.741	0.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.611	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	0.685	0.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646
40	0.681	0.851	1.050	1.303	1.684	2.021	2.123	2.423	2.704	2.971	3.307	3.551
50	0.679	0.849	1.047	1.299	1.676	2.009	2.109	2.403	2.678	2.937	3.261	3.496
60	0.679	0.848	1.045	1.296	1.671	2.000	2.099	2.390	2.660	2.915	3.232	3.460
80	0.678	0.846	1.043	1.292	1.664	1.990	2.088	2.374	2.639	2.887	3.195	3.416
100	0.677	0.845	1.042	1.290	1.660	1.984	2.081	2.364	2.626	2.871	3.174	3.390
1000	0.675	0.842	1.037	1.282	1.646	1.962	2.056	2.330	2.581	2.813	3.098	3.300
∞	0.674	0.841	1.036	1.282	1.645	1.960	2.054	2.326	2.576	2.807	3.091	3.291
	50%	60%	70%	80%	90%	95%	96%	98%	99%	99.5%	99.8%	99.9%
	Confidence level C											

F distribution for *p* = 0.05

For various pairs of degress of freedom (v_1,v_2), the tabled entries represent the critical values of F above which the proportion *p* = 0.05 of the distribution falls. (Example, for *df* = 3,8 and F = 2.92 is exceeded by *p* = 0.05 of the population.



***p* = 0.05**

	Degrees of freedom for the numerator v_1																	
v_2	1	2	3	4	5	6	7	8	9	10	12	15	20	30	40	60	120	∞
1	161.446	199.499	215.707	224.583	230.160	233.988	236.767	238.884	240.543	241.882	243.905	245.949	248.016	250.096	251.144	252.196	253.254	254.317
2	18.513	19.000	19.164	19.247	19.296	19.329	19.353	19.371	19.385	19.396	19.412	19.429	19.446	19.463	19.471	19.479	19.487	19.496
3	10.128	9.552	9.277	9.117	9.013	8.941	8.887	8.845	8.812	8.785	8.745	8.703	8.660	8.617	8.594	8.572	8.549	8.526
4	7.709	6.944	6.591	6.388	6.256	6.163	6.094	6.041	5.999	5.964	5.912	5.858	5.803	5.746	5.717	5.688	5.658	5.628
5	6.608	5.786	5.409	5.192	5.050	4.950	4.876	4.818	4.772	4.735	4.678	4.619	4.558	4.496	4.464	4.431	4.398	4.365
6	5.987	5.143	4.757	4.534	4.387	4.284	4.207	4.147	4.099	4.060	4.000	3.938	3.874	3.808	3.774	3.740	3.705	3.669
7	5.591	4.737	4.347	4.120	3.972	3.866	3.787	3.726	3.677	3.637	3.575	3.511	3.445	3.376	3.340	3.304	3.267	3.230
8	5.318	4.459	4.066	3.838	3.688	3.581	3.500	3.438	3.388	3.347	3.284	3.218	3.150	3.079	3.043	3.005	2.967	2.928
9	5.117	4.256	3.863	3.633	3.482	3.374	3.293	3.230	3.179	3.137	3.073	3.006	2.936	2.864	2.826	2.787	2.748	2.707
10	4.965	4.103	3.708	3.478	3.326	3.217	3.135	3.072	3.020	2.978	2.913	2.845	2.774	2.700	2.661	2.621	2.580	2.538
11	4.844	3.982	3.587	3.357	3.204	3.095	3.012	2.948	2.896	2.854	2.788	2.719	2.646	2.570	2.531	2.490	2.448	2.404
12	4.747	3.885	3.490	3.259	3.106	2.996	2.913	2.849	2.796	2.753	2.687	2.617	2.544	2.466	2.426	2.384	2.341	2.296
13	4.667	3.806	3.411	3.179	3.025	2.915	2.832	2.767	2.714	2.671	2.604	2.533	2.459	2.380	2.339	2.297	2.252	2.206
14	4.600	3.739	3.344	3.112	2.958	2.848	2.764	2.699	2.646	2.602	2.534	2.463	2.388	2.308	2.266	2.223	2.178	2.131
15	4.543	3.682	3.287	3.056	2.901	2.790	2.707	2.641	2.588	2.544	2.475	2.403	2.328	2.247	2.204	2.160	2.114	2.066
16	4.494	3.634	3.239	3.007	2.852	2.741	2.657	2.591	2.538	2.494	2.425	2.352	2.276	2.194	2.151	2.106	2.059	2.010
17	4.451	3.592	3.197	2.965	2.810	2.699	2.614	2.548	2.494	2.450	2.381	2.308	2.230	2.148	2.104	2.058	2.011	1.960
18	4.414	3.555	3.160	2.928	2.773	2.661	2.577	2.510	2.456	2.412	2.342	2.269	2.191	2.107	2.063	2.017	1.968	1.917
19	4.381	3.522	3.127	2.895	2.740	2.628	2.544	2.477	2.423	2.378	2.308	2.234	2.155	2.071	2.026	1.980	1.930	1.878
20	4.351	3.493	3.098	2.866	2.711	2.599	2.514	2.447	2.393	2.348	2.278	2.203	2.124	2.039	1.994	1.946	1.896	1.843
21	4.325	3.467	3.072	2.840	2.685	2.573	2.488	2.420	2.366	2.321	2.250	2.176	2.096	2.010	1.965	1.916	1.866	1.812
22	4.301	3.443	3.049	2.817	2.661	2.549	2.464	2.397	2.342	2.297	2.226	2.151	2.071	1.984	1.938	1.889	1.838	1.783
23	4.279	3.422	3.028	2.796	2.640	2.528	2.442	2.375	2.320	2.275	2.204	2.128	2.048	1.961	1.914	1.865	1.813	1.757
24	4.260	3.403	3.009	2.776	2.621	2.508	2.423	2.355	2.300	2.255	2.183	2.108	2.027	1.939	1.892	1.842	1.790	1.733
30	4.171	3.316	2.922	2.690	2.534	2.421	2.334	2.266	2.211	2.165	2.092	2.015	1.932	1.841	1.792	1.740	1.683	1.622
40	4.085	3.232	2.839	2.606	2.449	2.336	2.249	2.180	2.124	2.077	2.003	1.924	1.839	1.744	1.693	1.637	1.577	1.509
60	4.001	3.150	2.758	2.525	2.368	2.254	2.167	2.097	2.040	1.993	1.917	1.836	1.748	1.649	1.594	1.534	1.467	1.389
120	3.920	3.072	2.680	2.447	2.290	2.175	2.087	2.016	1.959	1.910	1.834	1.750	1.659	1.554	1.495	1.429	1.352	1.254
∞	3.841	2.996	2.605	2.372	2.214	2.099	2.010	1.938	1.880	1.831	1.752	1.666	1.571	1.459	1.394	1.318	1.221	1.000

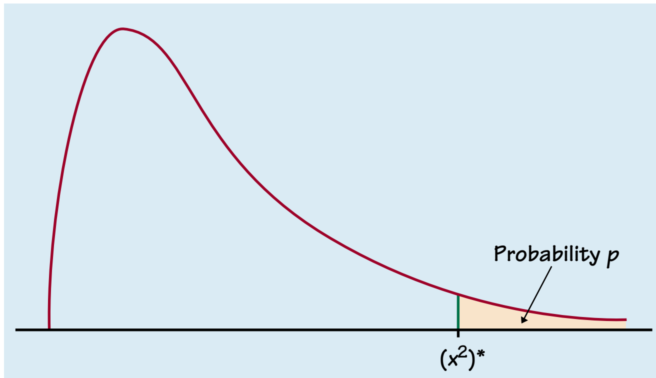


Table entry for p is the point $(X^2)^*$ with probability p lying above it.

TABLE F X^2 distribution critical values

	Tail probability p											
df	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.32	1.64	2.07	2.71	3.84	5.02	5.41	6.63	7.88	9.14	10.83	12.12
2	2.77	3.22	3.79	4.61	5.99	7.38	7.82	9.21	10.60	11.98	13.82	15.20
3	4.11	4.64	5.32	6.25	7.81	9.35	9.84	11.34	12.84	14.32	16.27	17.73
4	5.39	5.99	6.74	7.78	9.49	11.14	11.67	13.28	14.86	16.42	18.47	20.00
5	6.63	7.29	8.12	9.24	11.07	12.83	13.39	15.09	16.75	18.39	20.51	22.11
6	7.84	8.56	9.45	10.64	12.59	14.45	15.03	16.81	18.55	20.25	22.46	24.10
7	9.04	9.80	10.75	12.02	14.07	16.01	16.62	18.48	20.28	22.04	24.32	26.02
8	10.22	11.03	12.03	13.36	15.51	17.53	18.17	20.09	21.95	23.77	26.12	27.87
9	11.39	12.24	13.29	14.68	16.92	19.02	19.68	21.67	23.59	25.46	27.88	29.67
10	12.55	13.44	14.53	15.99	18.31	20.48	21.16	23.21	25.19	27.11	29.59	31.42
11	13.70	14.63	15.77	17.28	19.68	21.92	22.62	24.72	26.76	28.73	31.26	33.14
12	14.85	15.81	16.99	18.55	21.03	23.34	24.05	26.22	28.30	30.32	32.91	34.82
13	15.98	16.98	18.20	19.81	22.36	24.74	25.47	27.69	29.82	31.88	34.53	36.48
14	17.12	18.15	19.41	21.06	23.68	26.12	26.87	29.14	31.32	33.43	36.12	38.11
15	18.25	19.31	20.60	22.31	25.00	27.49	28.26	30.58	32.80	34.95	37.70	39.72
16	19.37	20.47	21.79	23.54	26.30	28.85	29.63	32.00	34.27	36.46	39.25	41.31
17	20.49	21.61	22.98	24.77	27.59	30.19	31.00	33.41	35.72	37.95	40.79	42.88
18	21.60	22.76	24.16	25.99	28.87	31.53	32.35	34.81	37.16	39.42	42.31	44.43
19	22.72	23.90	25.33	27.20	30.14	32.85	33.69	36.19	38.58	40.88	43.82	45.97
20	23.83	25.04	26.50	28.41	31.41	34.17	35.02	37.57	40.00	42.34	45.31	47.50
21	24.93	26.17	27.66	29.62	32.67	35.48	36.34	38.93	41.40	43.78	46.80	49.01
22	26.04	27.30	28.82	30.81	33.92	36.78	37.66	40.29	42.80	45.20	48.27	50.51
23	27.14	28.43	29.98	32.01	35.17	38.08	38.97	41.64	44.18	46.62	49.73	52.00
24	28.24	29.55	31.13	33.20	36.42	39.36	40.27	42.98	45.56	48.03	51.18	53.48
25	29.34	30.68	32.28	34.38	37.65	40.65	41.57	44.31	46.93	49.44	52.62	54.95
26	30.43	31.79	33.43	35.56	38.89	41.92	42.86	45.64	48.29	50.83	54.05	56.41
27	31.53	32.91	34.57	36.74	40.11	43.19	44.14	46.96	49.64	52.22	55.48	57.86
28	32.62	34.03	35.71	37.92	41.34	44.46	45.42	48.28	50.99	53.59	56.89	59.30
29	33.71	35.14	36.85	39.09	42.56	45.72	46.69	49.59	52.34	54.97	58.30	60.73
30	34.80	36.25	37.99	40.26	43.77	46.98	47.96	50.89	53.67	56.33	59.70	62.16
40	45.62	47.27	49.24	51.81	55.76	59.34	60.44	63.69	66.77	69.70	73.40	76.09
50	56.33	58.16	60.35	63.17	67.50	71.42	72.61	76.15	79.49	82.66	86.66	89.56
60	66.98	68.97	71.34	74.40	79.08	83.30	84.58	88.38	91.95	95.34	99.61	102.7
80	88.13	90.41	93.11	96.58	101.9	106.6	108.1	112.3	116.3	120.1	124.8	128.3
100	109.1	111.7	114.7	118.5	124.3	129.6	131.1	135.8	140.2	144.3	149.4	153.2